

Thesis Talk

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Overview of Topics

- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Discussion
- Outlook

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- At the beginning, oversample and knit an entire top because the run time was too long
- Trying to make an accurate model for crack propagation in snow
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will probably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:

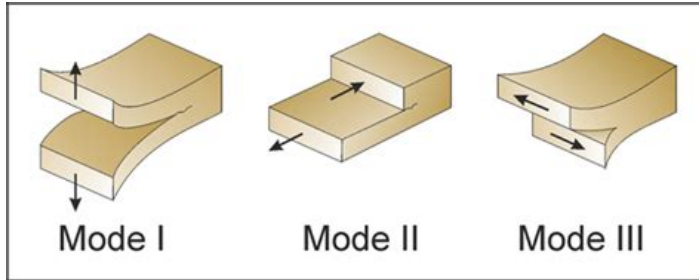


Figure: Fracture modes according to Philipp et al., 2013

Words I will probably say a lot explained II

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (Gaume et al., 2018)
- Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow (Trottet et al., 2022)
- Supershear: Cracks propagate faster than shear wave speed in snow (Trottet et al., 2022)

Overview of Models

- Model domain 400m wide, 3m high
- Absorbing boundary conditions 30m thick along top and both sides of domain
- No ABC at bottom because of theoretical snow interface
- 1m snow layer, 2m air layer
- Multiple source versions: single source, source line with sources firing with a time delay (moving source) and rupture front
- Sources buried in snow (like crack)
- For moving source: sub-rayleigh and supershear speeds
- 10Hz central frequency ricker source wavelet

My latest creation: the cohesive zone model (CZM)

- Model based on the cohesive zone model by Mahajan and Joshi, 2008
- In CZM, fracturing is a gradual process, separation of the crack faces takes place over an extended crack tip, the cohesive zone (Mahajan & Joshi, 2008)
- Resistance to opening by cohesive traction (Mahajan & Joshi, 2008)
- In other words: there is a zone ahead of the crack tip, which already has some damage of the bonds
- Cracks represented as a fraction between different crack modes (Mode I-III)

Bibliography I



Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018). Dynamic anticrack propagation in snow. *Nature Communications*, 9(1), 3047.
<https://doi.org/10.1038/s41467-018-05181-w>



Mahajan, P., & Joshi, S. K. (2008). Modeling of interfacial crack velocities in snow. *Cold Regions Science and Technology*, 51(2), 98–111.
<https://doi.org/10.1016/j.coldregions.2007.05.008>



Philipp, S., Afşar, F., & Gudmundsson, A. (2013). Effects of mechanical layering on hydrofracture emplacement and fluid transport in reservoirs. *Frontiers in Earth Science*, 1. <https://doi.org/10.3389/feart.2013.00004>



Schweizer, J., Bruce Jamieson, J., & Schneebeili, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123>



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.

Bibliography II



Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, 18(9), 1094–1098.
<https://doi.org/10.1038/s41567-022-01662-4>



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